

The Impact of Physical Activity on Mental Well-being and Stress Levels

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Introduction:

Physical activity is widely recognized as a cornerstone of both physical health and overall psychological stability. For students and young adults, maintaining a consistent exercise routine can be a significant challenge due to academic demands, sedentary lifestyle habits and high screen times. However, the physiological release of endorphins provided by physical activity can be vital for managing the pressures of modern life.

This report explores the correlation between physical activity levels and mental well-being using both qualitative and quantitative data analysis. By surveying a sample of 100 individuals, factors like workout frequency, activity intensity and even sleep quality will be examined in order to uncover patterns that reveal the broader impact of a physically active lifestyle. This study will provide valuable insights for individuals looking to optimize their mental health through movement and for institutions aiming to support student well-being through sports and wellness programs.

Research Question:

What is the relationship between physical activity levels and mental well-being among young individuals?

Hypothesis:

It is hypothesized that individuals who engage in physical activity more frequently will report significantly higher levels of happiness and productivity, alongside lower stress levels, compared to those who live sedentary lifestyles. This hypothesis is built on the biological premise that regular exercise regulates cortisol, the stress hormone, and enhances mood through increased neurotransmitter activity.

Methodology:

Population of Interest: The population of interest for this study consists of students and working professionals, ranging from young adults to those in their late twenties or thirties. While the study includes a significant number of university students, the sample captures a broader demographic of individuals navigating the challenge of balancing daily productivity with physical and mental well-being.

Sampling Method: *Convenience sampling*, a method where participants are selected based on accessibility and proximity, was the sampling technique used for this report. Through a digital survey uploaded to WhatsApp groups and other forms of social media, a sample of **100** individuals, varying from under 18 to over 24, was obtained. This method was selected for its cost-effectiveness and ability to gather a high volume of responses in a limited time. While this may limit the diversity of the sample, it remains a highly effective approach for this particular research project. To ensure better data quality, the form was shared across different academic departments, and further shared among families and friends, increasing the variability of the respondent profile.

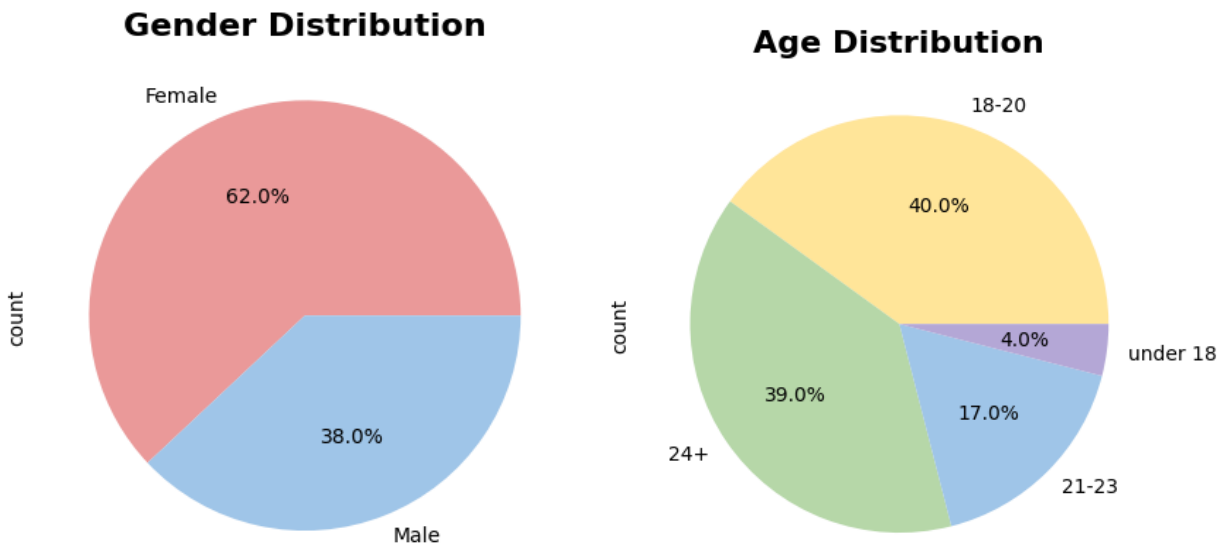
Bias Identification: *Self-reporting bias* was identified as a primary factor in the design of this research. Participants may naturally overestimate their physical activity levels or underestimate their stress levels in order to align with perceived social norms. To reduce the effect of this bias, we ensured complete anonymity for all respondents, encouraging more honest self-assessment. Additionally, we used a 1-10 scale for emotional metrics to provide a neutral framework for evaluation. Moreover, the survey avoided leading questions, primarily focusing on straightforward wording and descriptive categories for exercise, ensuring that the data would capture variety without influencing the respondent's choice.

Survey Link: <https://forms.gle/nNQuX7oRktx4Ja1WA>

Survey Questions

- What is your age group?
- What is your gender?
- How many days per week do you engage in physical activity?
- How would you describe your overall physical activity level?
- What type of activity do you do most often?
- On average, how long is each session?
- Which best describes your daily movement? (outside of exercising)
- On a scale of 1–10, how happy do you feel overall?
- On a scale of 1–10, how stressed do you feel on average?
- On a scale of 1–10, how would you rate your overall mental well-being?
- How many hours do you sleep per night?
- How would you rate your sleep quality?
- How many hours do you spend on screens daily?
- On a scale of 1–10, how energetic do you feel on a typical day?
- On a scale of 1–10, how productive do you feel during the day?
- On days you're active, how does your mood change?
- Do you believe physical activity improves mental health?

Demographic Overview:



The survey reached a diverse group of 100 participants. While the study initially targeted students, the data shows that 39% of respondents are aged 24 or older. This variety allows the research to reflect how exercise affects well-being across different stages of early adulthood, rather than just the typical college age bracket.

Additionally, the sample is composed of 62% females and 38% males. This distribution provides a comprehensive look at mental health across genders, with a particularly strong insight into the habits of female participants. The diversity in both age and gender ensures that the study's findings are representative of a broad spectrum of personal experiences and lifestyle choices.

Happiness Score (1-10):

Statistics:

- **Mean:** 6.18
- **Median:** 6.0
- **Mode:** 7.0

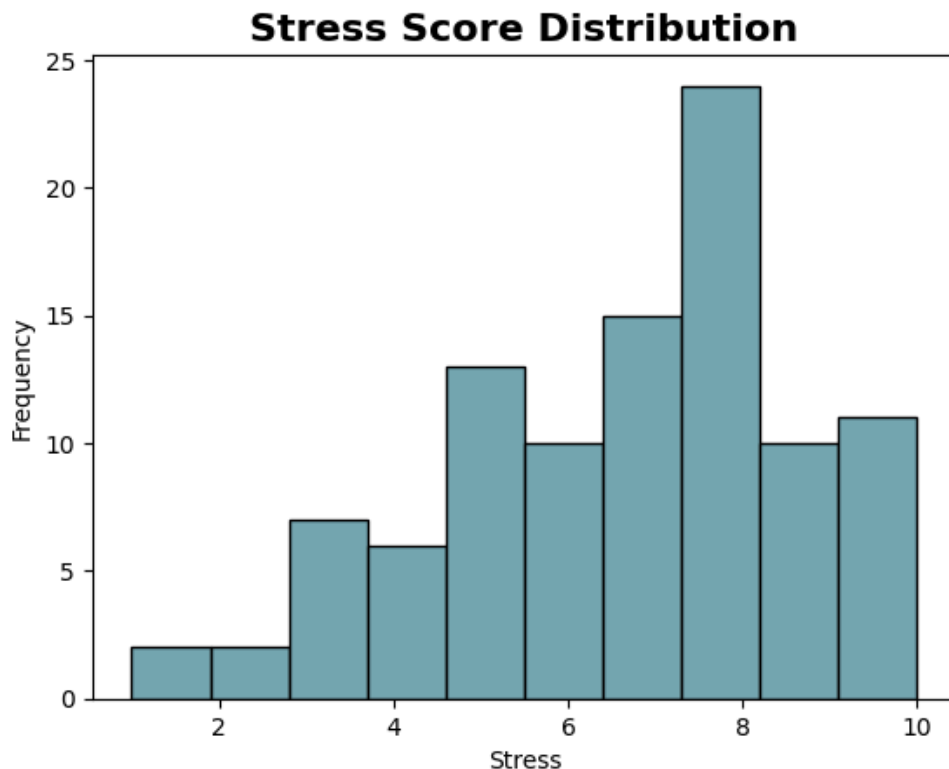


The distribution of happiness among the **100** individuals is **bimodal**, featuring two prominent peaks. The highest frequency occurs in the **7.0** range, with a secondary significant peak in the **5.0** range. The graph also shows a **slight left skew**, suggesting an overall positive sentiment across the group. The dip between the two peaks (near the 6.0 mark) could imply **two distinct subgroups** within the sample: one that is "content" and another that is "highly satisfied."

Stress Score (1-10):

Statistics:

- **Mean:** 6.73
- **Median:** 7.0
- **Mode:** 8.0



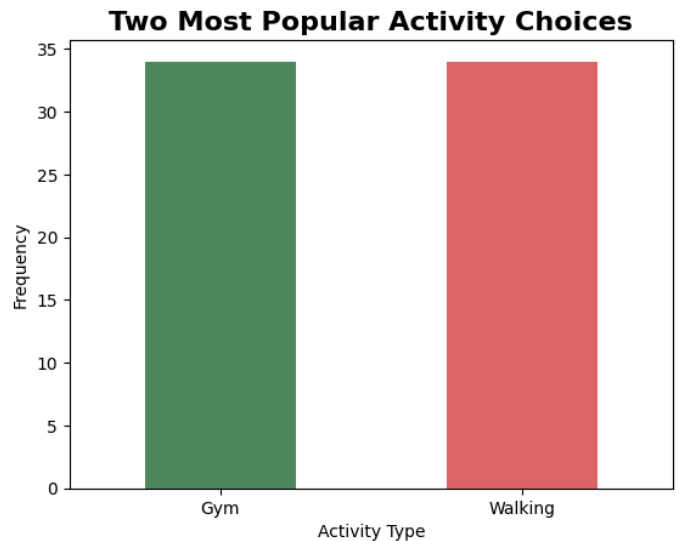
The distribution is notably **skewed to the left**, with a very strong peak (mode) in the **8.0 range**, reaching a frequency of 24. Unlike the previous happiness graph which felt more balanced, this data is heavily weighted toward the higher end of the scale. The data reveals a **high-stress environment**, as the most frequent score is an 8 out of 10. This could suggest a sample that is high-performing or highly active, but potentially deals with other stressful activities.

Descriptive Statistics of Well-being Indicators

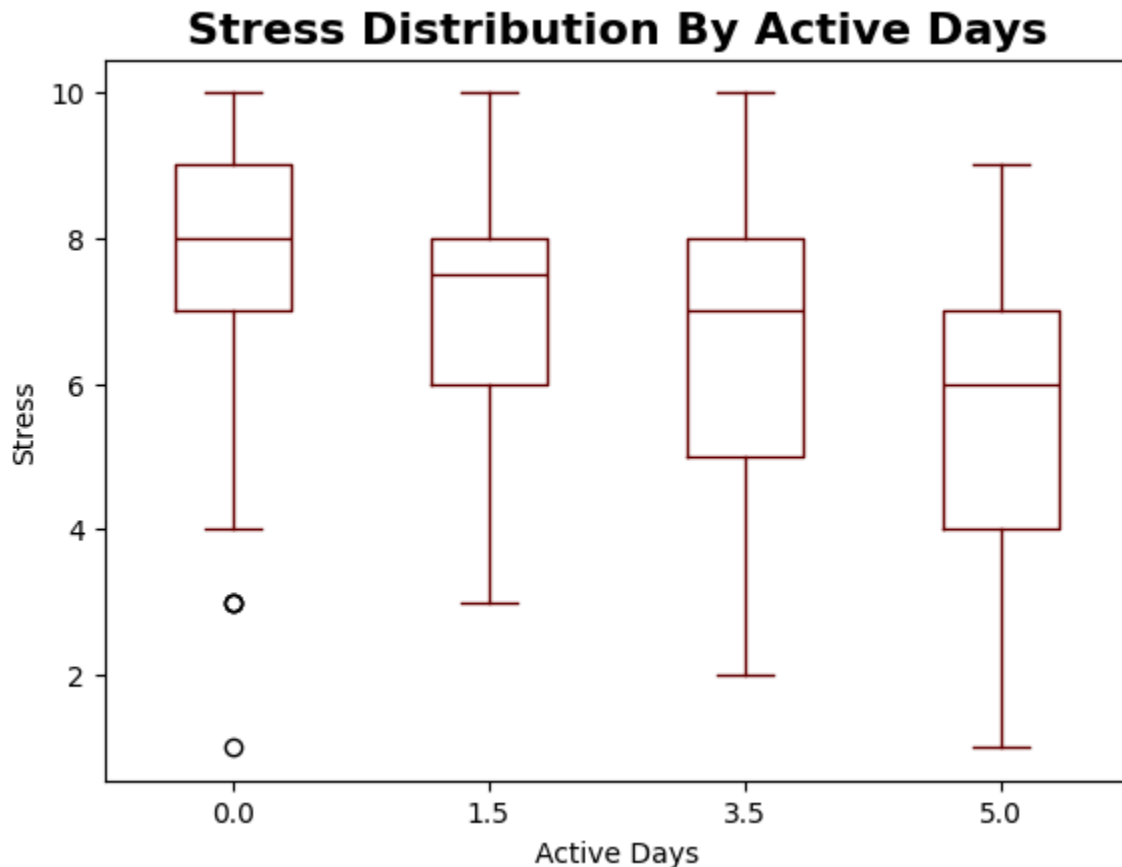
Indicator	Mean	Median	Mode
<i>Happiness</i>	<i>6.18</i>	<i>6.0</i>	<i>7</i>
<i>Stress</i>	<i>6.73</i>	<i>7.0</i>	<i>8</i>
Well-being	6.2	6.5	5
Productivity	6.14	6.0	7
Energy	5.88	6.0	7

Most Common Choices of Physical Activity:

This bar chart shows an **exact tie** between the two most popular activities. Both "**Gym**" and "**Walking**" reached a frequency of **34**, each being a third of the sample. This balance suggests that the population values physical activity, though they are divided on the preferred environment and intensity level, with preference between high-intensity indoor exercise (Gym) and low-intensity outdoor/steady-state exercise (Walking).

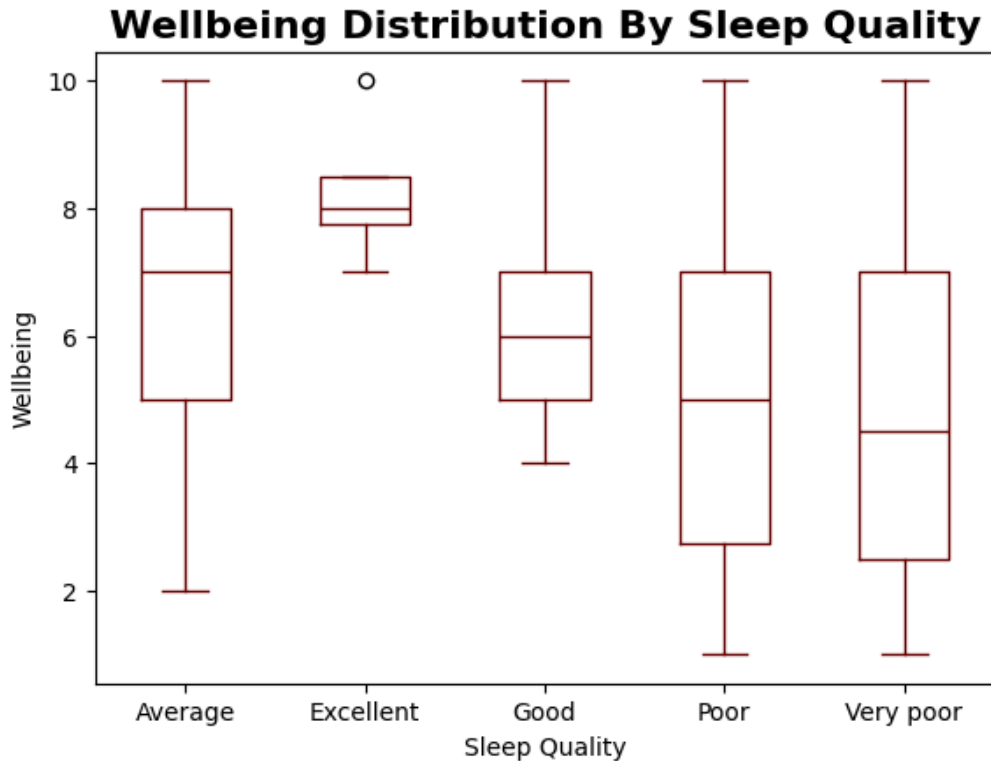


Boxplot of Stress Distribution by Average Active Days:



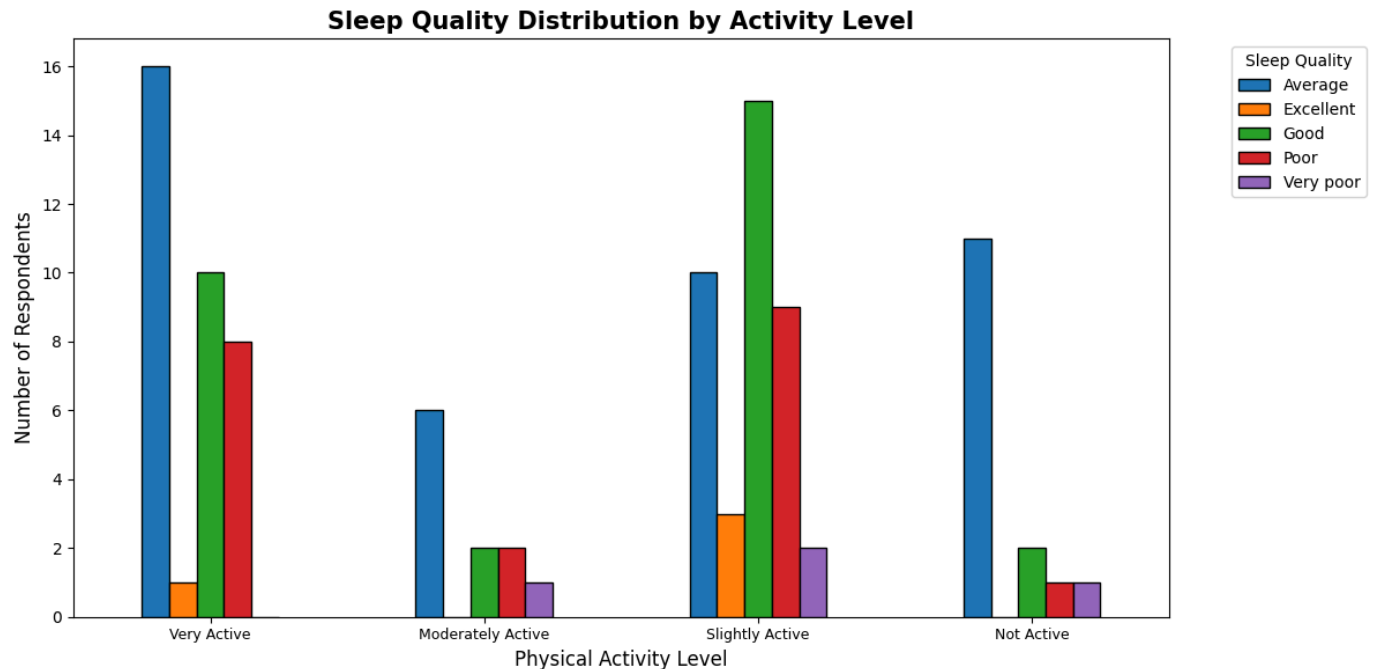
There is a clear **inverse relationship** between the number of active days and stress levels. As the number of active days increases from 0.0 to 5.0, the **median stress score** (the horizontal line within each box) steadily drops from 8.0 down to 6.0. The graph suggests that **regular physical activity acts as a significant buffer against stress**. Moving from zero activity to 5 days of activity per week corresponds with a **25% reduction** in median stress levels.

Boxplot of Wellbeing Distribution by Sleep Quality:



There is a strong, direct link between sleep quality and overall wellbeing. The **median wellbeing score** peaks sharply for those with "Excellent" sleep (at an 8.0) and declines consistently as sleep quality worsens, bottoming out for the "Very poor" group (at approximately 4.5). The data demonstrates that **sleep quality is a primary driver of wellbeing**. While high-quality sleep (Excellent) appears to guarantee a high level of wellbeing, poor sleep quality doesn't just lower wellbeing but makes it much more inconsistent.

Comparing Sleep Quality of Each Activity Level

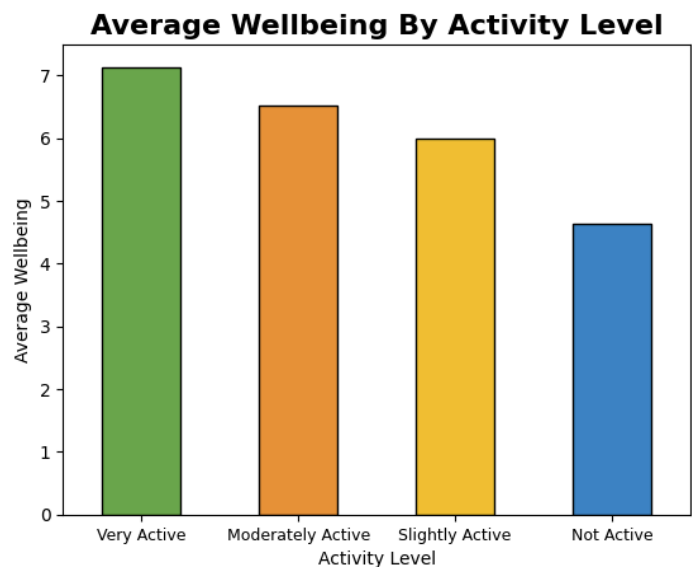


This multibar chart illustrates a complex and somewhat non-linear relationship between activity levels and sleep quality among respondents. A striking observation is that “Average” sleep quality is the most frequent response across almost all activity levels, particularly peaking among the “Very Active” group. Interestingly, the “Slightly Active” group appears to have the highest frequency of “Good” sleep, surpassing even the most active individuals. Moreover, the “Very Active” group shows a high count of “Poor” sleep, suggesting that while staying active generally supports sleep, **extreme activity might not always translate to better rest.**

According to the previous chart along with this one, the relationship between activity and wellbeing is indirect and somewhat dependent on sleep, meaning that while activity boosts wellbeing by improving sleep quality, **excessive activity** can actually **undermine wellbeing** by disrupting the necessary rest to sustain it.

Average Wellbeing by Activity Level

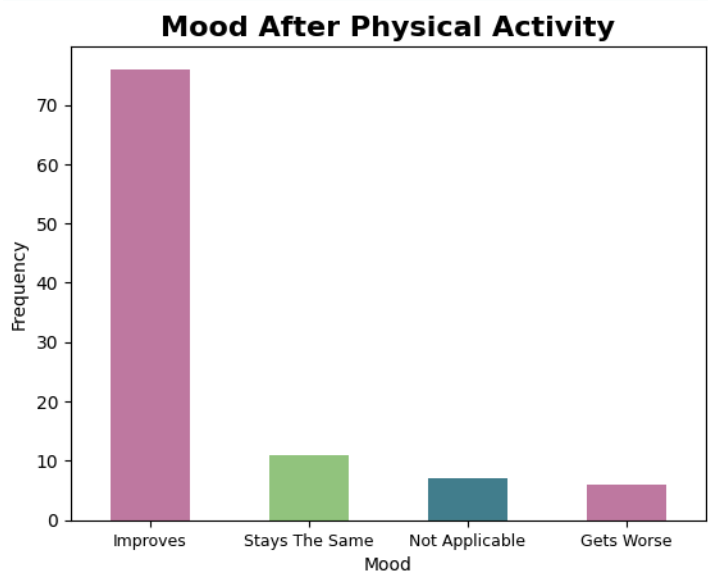
The chart displays a perfect step-down trend, showing that as physical activity decreases, **average wellbeing** follows suit. The most substantial gap occurs between being "Slightly Active" and "Not Active." This suggests that even a small amount of movement provides a meaningful boost to wellbeing compared to total inactivity. As a result, the data provides a compelling case for the benefits that accompany higher levels of activity.



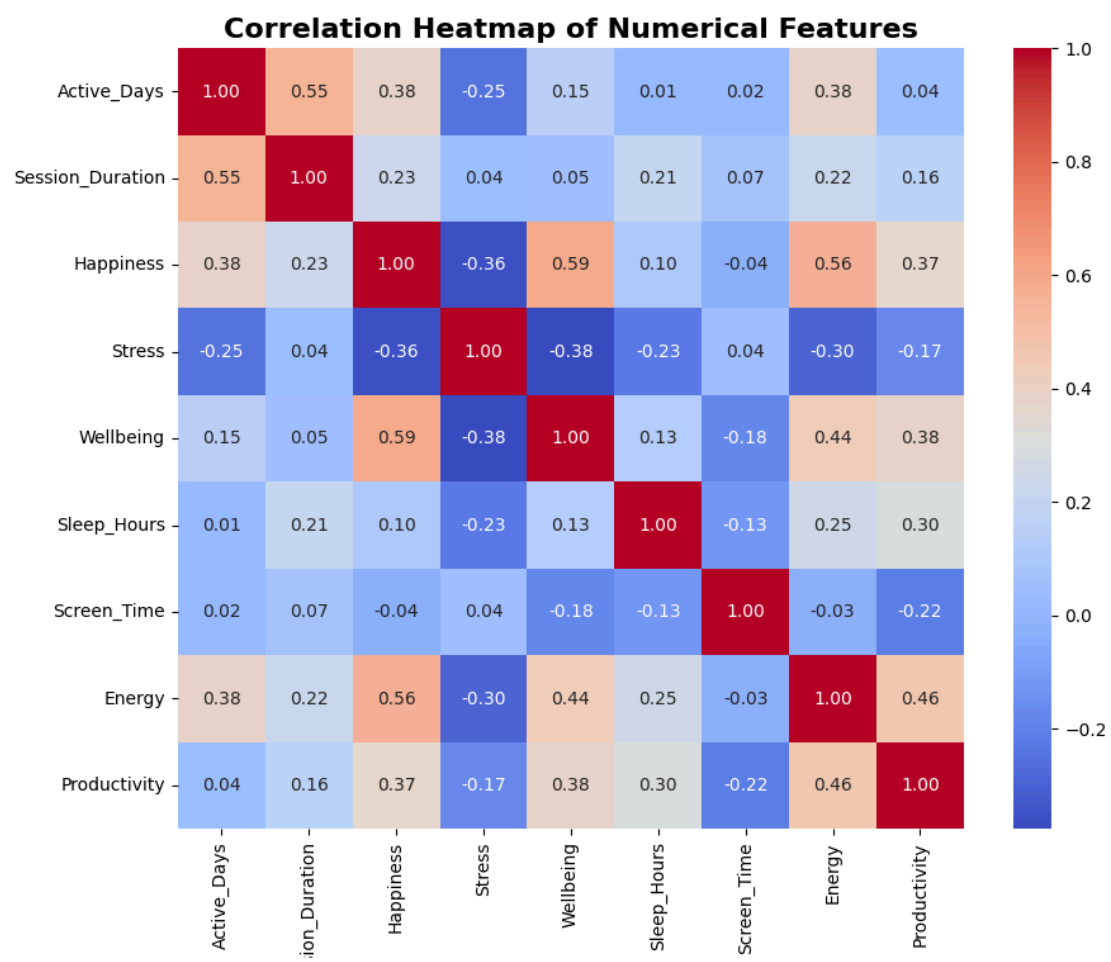
Mood Change After Physical Activity

The results are overwhelmingly positive, with the **"Improves"** category reaching a frequency of approximately **76**. This dwarfs all other responses combined, indicating a nearly universal psychological benefit to exercise

within this group. A small portion of respondents report that their mood **"Stays The Same"** or find the question **"Not Applicable"** (likely those from the "Not Active" group identified in previous graphs). The data provides definitive evidence for the **immediate mood-boosting effects** of physical activity.



Final Correlation Heatmap for All Variables Considered



The heatmap reveals a **wellness cluster** consisting of happiness, wellbeing, and energy. If you improve one, the others are very likely to rise. On the flip side, **stress** acts as the primary antagonist to this cluster. This suggests that aspects of mental health tend to improve together rather than independently.

In contrast, stress exhibits consistent negative correlations with nearly all variables. Physical activity variables, such as active days and activity level, show moderate positive correlations with the wellness cluster and negative correlations with stress, indicating a supportive yet indirect role. Overall, the heatmap suggests that mental health is influenced by an interconnected system of lifestyle factors.

Conclusion:

The results of this study provide a general support for the original hypothesis: an important relationship between physical activity levels and mental well-being exists. The data has shown a consistent positive relationship between activity levels and well-being indicators such as happiness, energy, productivity and overall well-being. At the same time, higher levels of physical activity are associated with lower stress levels, particularly evident in the boxplot comparing stress across active days, where stress steadily decreases as activity increases.

However, the relationship is not perfectly linear or isolated. Some findings have suggested that additional factors, especially sleep quality, play a significant mediating role between physical activity and mental health. For instance, while higher activity levels generally improve well-being, excessively high levels may be linked with poorer sleep quality in some cases, which can negatively affect mental stability.

All in all, the hypothesis is largely supported: individuals who are more physically active tend to report better mental well-being and lower stress levels than those who are sedentary. However, the data also suggests that physical activity is one of the various interacting lifestyle factors that affect mental health rather than a sole determinant.

Critical Reflection:

- **Convenience sampling bias:** may limit diversity and make sample less representative
- **Self-reporting bias:** personal perception may cause over/underestimation of data
- **Somewhat subjective scales:** Measures like “Good sleep” or “Moderate activity” may have been interpreted differently by each respondent, leading to data inconsistencies
- **Demographic imbalance:** Higher proportion of females and young adults rather than a balanced distribution
- **Correlation VS causation:** Other factors may play a role, therefore, relationships identified between variables cannot fully prove that physical activity is what directly causes improved mental well-being.